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RESEARCH ARTICLE

Speaker-Independent Phoneme-Based Automatic Quranic Speech Recognition Using Deep Learning

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ABSTRACT An automatic speech recognition system is important to help Muslims recite the Holy Quran accurately. Most existing research ignores a wide range of potential users (reciters) in their systems by focusing on professional adult male reciters due to the abundance of this group's recordings and the lack of annotated data for other groups. This work bridges this gap by developing a speaker-independent system that recognizes Quranic recitations of different genders, ages, accents, and Tajweed levels. Our recognizer is designed on the phoneme level to offer Tajweed detection. Using a private dataset, rich of non-transcribed recitations, we propose training the DeepSpeech model with Transfer Learning and semi-supervised learning techniques. The performance of our model is evaluated using several proposed language models and evaluation metrics, including Word Error Rate (WER) and Phoneme Error Rate (PER). The goal is to show how our model would perform in regard to diverse reciter groups. Starting with a typical test set of unseen professional adult male recitations, the WER/PER of our model are 3.11% and 6.18%, respectively. More interestingly, our model achieves a WER of 25.39% and 17.93% when tested on recitations of non-professional (normal) females and children, respectively. The results are very promising and ensure the ability of our model to recognize recitations of various groups of normal reciters. Moreover, the latter results were done on the public "in-the-wild" Tarteel dataset, hoping this will be useful for comparison with future research and building more practical recitation teaching applications. In fact, a major limitation of existing systems (including ours) is the ability to handle diverse in-the-wild scenarios, such as when the reciter is reciting the verses in a very high tempo (common for those trying to memorize the Quran.

INDEX TERMS Arabic, deep learning, DeepSpeech, language model, natural language processing, Quran, semi-supervised learning, speech recognition, transfer learning.

I. INTRODUCTION

The Holy Quran contains the divine words of ALLAH (the one true GOD), revealed to Prophet Mohammad (peace be upon him) through the angel Gabriel over a span of 23 years. The Holy Quran is important for all Muslims and essential in their daily lives. Reading the Quran is required for performing Islamic daily prayers. There are more than 1.5 billion Muslims around the world including a large percentage of non-native Arabic speakers [1]. Muslims are instructed to read the Quran as it was revealed, following recitation rules (Tajweed). Without the presence of a teacher, Automatic Speech Recognition (ASR) systems are emerging as a crucial tool to help Muslims recite the Quran accurately. This is of great importance for non-native Arabic speakers, Muslims in non-Muslim countries, kids, and the visually impaired. Handling a wide range of users in such systems is very challenging. Most of the existing Quranic ASR systems are trained and evaluated on recitations by professional adult male reciters. In fact, such models are most likely to perform poorly on other groups, such as females, children, and non-professional "normal" reciters. Indeed, available systems are very limited and do not fulfill the needs of the Muslim community.

Speech recognition has been an area of interest for researchers around the world for many years with

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applications across different domains [2], [3], [4]. Deep learning (DL), which is a type of machine learning (ML), involves the use of neural networks (NN) to model and solve complex problems as in [5] and [6]. Recent developments in Deep Learning have led to remarkable improvements in speech recognition results over those of traditional models [7]. Various types of DL-based architectures exist for ASR. For instance, DeepSpeech is a Connectionist temporal classification (CTC)-powered model that has achieved excellent performance on benchmark speech datasets of different languages [8]. Despite all DL advantages in ASR, it typically requires a large amount of data and computing resources to achieve good results [9].

The Quran suffers from a lack of speech recognition data resources. Indeed, collecting manually transcribed speech data from many speakers is costly and time-consuming. One of the key advancements in DL to overcome the data limitations is Transfer Learning. It involves leveraging knowledge gained from one task into another related task with limited data [10]. Furthermore, semi-supervised learning is a popular technique used to overcome the lack of transcribed data issue.

Our objective is to build a Quranic ASR system that serves Muslims of all genders, ages, fluency levels, and Arabic accents. This work proposes a speaker-independent continuous speech recognition system for Quranic recitations based on DeepSpeech, which is typically trained in a supervised manner. In fact, we proposed to start with supervised training as the baseline. Whilst, the final model is developed using transfer learning followed by semi-supervised training. We design the recognizer to be phoneme-based to additionally support Tajweed detection. That is, our model automatically transcribes Quranic recitations into a written sequence of phonemes representing standard Arabic and Tajweed sounds. We aim to make this model available for use in digital applications that facilitate Quran learning. Such a system can be beneficial for Tajweed checking and correction, memorizing the Quran's verses, and Quran search by voice. Since Quran is a special kind of Arabic, our model can also be utilized for Arabic pronunciation recognition and correction.

In summary, the main contributions of this work are as follows:

- Adopting phoneme-based DeepSpeech for recitation errors detection, at the level of words and Tajweed rules, which can serve various digital Quran applications.
- Using transfer learning from English to Quran to improve Quranic model performance, focusing on possible gender bias introduced by unbalanced speech data used.
- Introducing a semi-supervised method to support the system with reliable high-confidence pseudo-labeled training data, rich in females and children recitations.
- Utilizing the publicly available Tarteel dataset to evaluate speaker-independence in further Quranic recognition research.

The rest of this paper is organized as follows: Section II provides background and literature review. Section III describes the used datasets. The proposed phonemes, model architecture, training, and post-processing methods are presented in Section IV. Section V summarizes experiments, results, and analysis. Finally, conclusions and future work are provided in Section VI.

II. BACKGROUND AND LITERATURE REVIEW

The Holy Quran is divided into 114 chapters of unequal size, where each chapter is called *Surah* in Arabic. Surah is further divided into verses (*Ayah*s in Arabic). The whole Quran has a few thousand unique words therefore considering it as a limited but large vocabulary corpus. The Quran is also divided into 30 roughly equal sections, each of which is called *Juz'*s in Arabic.

The Holy Quran was revealed in the classical Arabic language. Arabic has 29 letters in its alphabet (including *Hamza*). Letters alone do not provide enough information about the correct pronunciation, thus, Arabic text provides many *diacritics* to guide the reader. Quran can be written in different styles according to the amount of diacritics and symbols. Most of the current printed Quran versions are written in a special style called **Uthmani** [11]. Digital Quran can also be written in the **Imla'ei:** style. One example to illustrate the difference between different styles is shown in Table 1. The whole Unicode text of the Quran in its several types can be downloaded from the Tanzil project website.[1]

There are ten different schools available for recitations of the Quran which is known in Arabic as *Qira'at*. The differences between them are very slight and related to altering some diacritics of the letters and sometimes the letters themselves.[2]

The followers of Islam are ordered to recite the Quran with Tajweed. In linguistic terms, Tajweed is "improvement". According to [12], the applied definition is: "articulating every letter from its articulation point and giving the letter its rights and dues of characteristics". In general, the pronunciation of Arabic words may change according to their position within other words of the spoken sentence. Tajweed adds some pronunciation rules that apply to letters in some situations during Quran recitation only. Said otherwise, mistakes in Tajweed can be divided into two levels. First, the clear mistake that occurs by dropping one word, letter, or vowel, changing anyone into another, etc. This kind of mistake affects the accuracy of the reading but is obvious and easily noticeable by anyone who knows Arabic. Alongside, there is the hidden mistake type which happens when missing a Tajweed rule. Unlike clear mistakes, hidden mistakes do not change the meaning of the speech. Indeed, only masters of the Quran reading can detect it [12].

Many research efforts have been conducted for the Holy Quran using conventional ASR tools, mostly based on Hidden

[1] http://tanzil.net
[2] https://en.wikipedia.org/wiki/Qira'at

TABLE 1. Example of different digital formats of Quran text.

Format	Example	Format		Example
Uthmani	إِذْ نَادَىٰ رَبَّهُۥ نِدَآءً خَفِيًّا	Imla'ei	Simple	إِذْ نَادَى رَبَّهُ نِدَاءً خَفِيًّا
			Minimal	إِذ نادى رَبَّهُ نِداءً خَفِيًّا
			Clean	إذ نادى ربه نداء خفيا

Markov Model (HMM). Reference [13] has published a detailed review of conventional Arabic ASR in [13]. HMM is a statistical model that uses a Markov process that contains hidden states. It is used to estimate the probability of phonemes or word sequences given a specific observed sequence of speech feature vectors. The systems published in [14], [15], [16], [17], [18], and [19] used Sphinx tool for Quranic verses recognition (QVR). Similarly, the QVR system was built using PocketSphinx [20]. Otherwise, the work in [21] used the Hidden Markov Model Toolkit (HTK) to build a speaker-independent model using HMM with Gaussian mixtures. Apparently, the recognition performance depends on the HMM parameters, transcriptions format, phonemes list, and the used pronunciation dictionary.

Pronunciation dictionary is a core component of conventional ASR systems. Hyassat and Abu Zitar [18] have developed a toolkit for automatically generating both standard Arabic and Holly Quran pronunciation dictionaries based on hand-derived rules.

Some research on Tajweed checking systems has been published in the literature. Authors of [22] have built a model for recitation evaluation. The model can detect common Arabic pronunciation mistakes in Quranic recitations and, correspondingly, asks for correction. For this study, eight letters only were selected to validate the proposed work. The authors introduced a semi-automatic segmentation procedure to detect the boundaries of the target sounds of letters of interest. Then, pronunciation verification was obtained by extensive comparisons between test segments and six pre-recorded recitations from famous readers (including correct and most common recitation errors). In the same way, authors of E-hafiz [23] have developed an intelligent system to identify the Tajweed mistakes on word level. The system extracts features of recorded voices Mel-Frequency Cepstral Coefficient (MFCC) technique. Any mismatch between features compared to experts' voices stored in the system's database is considered a mistake. Model recognition accuracy was evaluated for 10 speakers for each group of men, children, and women, where each speaker recited some specific Ayah of the Holy Quran. Likewise, the SmarTajweed in [24] considered Tajweed rules recognition as a classification task and used Support Vector Machine (SVM) with the threshold scoring to recognize four different Tajweed rules.

Authors of [25] have developed a system for the Holy Quran memorization tutoring using Matlab. Surprisingly, the work was done based on Artificial Neural Network (ANN), instead of HMM, for acoustic modeling. The network is simply a fully connected Multi-Layer Perceptron (MLP) with one hidden layer of 8 neurons. Rather than building a speech-to-text model for Ayah recognition, the authors trained their proposed model to recognize the recited Ayah by Ayahs classification. To recognize mistakes of missing words, they trained their network on two classes: fully correct Ayah or missing words.

One of the limitations of conventional ASR models is the time alignment problem. That is, the training dataset should have the transcript units (words or phonemes) synchronized with speech segments. Manual alignment of training datasets by experts is the most accurate way for this purpose, however, it is very time-consuming. Instead, time-aligned transcriptions are derived automatically using forced Viterbi alignment in such conventional models to be able to perform typical training. Integrating alternative automatic time-alignment techniques also exists in the literature. However, the quality of the ASR acoustic model is affected by the quality of the alignments [26].

In recent years, some DL-based models have appeared to address Quranic ASR systems to overcome the limitations of conventional models. Published review papers of ASR for the Arabic language including the Quran can be found in [27] and [28]. For instance, a speech recognizer was introduced in [29] taking into account Tajweed rules. The system used Time Delay Neural Networks (TDNN) from the KALDI toolkit which showed the best performance results among other experiments.

Furthermore, some research has applied ASR end-to-end transformer-based models for the Holy Quran. Authors of [30] proposed a model that consists of a Convolutional Neural Network (CNN) encoder and a character-based decoder. They have trained their model on a dataset that covers about 37 chapters of the Quran. Similarly, the work in [31] used multi-head attention for the encoder and decoder.

DeepSpeech is a promising CTC-based deep learning model and it has been used for Quranic recognition in [32], [33], and [34]. Authors of [32] developed a model called DeepSpeech-Quran and made it available for developers. The work introduced a two-stage training process and data augmentation techniques to enhance the recognition of normal users. Likewise, the work in [33] addressed the poor performance of DeepSpeech trained on male recitations when tested for female reciters. As a solution, authors adopted benchmark datasets consisting of recitations from different gender and age groups for enhancing recognition

performance. The dataset was recently published and called QRFAM [34]. The female dataset was manually collected and transcribed from semi-professional Arab reciters. Unfortunately, the female records cover a subset of the Holy Quran. A summary of Quranic ASR approaches found in the literature is illustrated in Figure 1.

The majority of Quranic models were trained on professional recitations collected freely from the internet. These recitations are mostly intended for male reciters. Besides, most of the models were developed using small datasets consisting of short verses covering only a few chapters of the Holy Quran. Moreover, Quran ASR researchers mostly used private datasets with no benchmark comparison. Alsharhan and Ramsay [35] have presented a thorough investigation of the properties of data that affect the performance of Arabic ASR systems, including gender, dialect, and training size. Our work comes to fill in some of these gaps by proposing an effective speaker-independent ASR model. As illustrated in Section V, the scale of our work, its novel aspects and its impressive results allows it to stand out among state-of-the-art systems.

III. DATASETS

A. EqraTech DATASET

The dataset used in this work is derived from a private database owned by EqraTech company.[3] EqraTech database contains a large number of collected recitations of professional reciters and ordinary Muslims around the world. The database has two groups based on the availability of labeling information: labeled and unlabeled. The provided unlabeled audios are about one million, whereas almost a quarter of that amount is found to be labeled. Each audio in the labeled dataset has a corresponding Quranic transcription that was manually labeled by Arab Muslims. The labeled dataset mostly includes professional recitations recorded in high-quality environments. A few samples of this dataset have intentional recitation errors. In contrast, the unlabeled dataset contains recordings in real environments from more diverse ordinary reciters.

The original labeled database was provided in Carnegie Mellon University (CMU) Sphinx database format.[4] An illustration for database files can be found in Table 2. The *.fileids* file specifies a list of all audio paths. Whilst, the *.dic* file represents the pronunciation dictionary which was automatically generated as stated in [36]. Each word in the dictionary may have multiple pronunciation variants, represented as a sequence of phonemes. The dictionary includes different correct recitations (according to *Hafs*[5]) in addition to a few common erroneous recitations. Besides, the *.transcription* contains transcription texts for each audio file, written in the Uthmani style and mentioning pronunciation variants for each word.

[3] http://eqratech.com
[4] https://cmusphinx.github.io/wiki/tutorialam/#data-preparation
[5] Hafs is one of the Qira'at of the Quran.

The raw database format and structure are not suitable for modern ASR. First of all, we prepared our dataset by reformatting it into comma separated value (CSV) format. Next, we list the functions involved in our preparation and cleaning of the EqraTech dataset:

1) Preparing the first version of the dataset in CSV, involving *wav_filename*, *speaker*, *wav_duration* and *wav_filesize* attributes.
2) Transforming raw pronunciation dictionary to reflect our proposed phonemes list.
3) Generating phonetic transcription (sequence of phonemes) in conjunction with the pronunciation dictionary for the *transcript* attribute.
4) Finding and removing duplicate samples.
5) Removing samples with missing or corrupted audio.
6) Generating a consistent audio format and encoding, giving 16-bit mono audios of *.wav format with a 16 kHz sample rate.
7) Sorting in ascending order by file size.
8) Removing too long samples, considering a threshold of 24 seconds.
9) Removing non-speech samples detected using voice activity detector (VAD), considering empty or less than 50% of speech.

The clean labeled dataset of EqraTech consists of 267,194 recitation samples. It has ∼547 hours of samples (7.37 seconds on average) distributed between 90 different speakers. Speakers can be grouped into three levels in terms of record amount. Some of them have records of the entire Holy Quran while others have medium or a few samples. Surprisingly, there is only one female reciter and no children among the speakers. On the other hand, the clean unlabeled dataset has 879,578 samples (∼1608 hours). These samples involve very diverse ∼62K speakers of wide (unknown) distribution of genders and ages.

In this work, we proposed to split our labeled dataset into 80:10:10 holdout partitions with non-overlapping speakers. Specifically, we held out 10% of the data for development (dev) and another similar for testing (test) while the rest for training (train).

However, due to the obvious difference in sample amounts across speakers, this was challenging. Thus, we first grouped speakers into three K-mean clusters by their sample count. We were expecting 26.7K samples for each of the dev and test subsets. Subsequently, we approximately reached this number by randomly selecting 1, 4, and 6 speakers from the largest group to the smallest. Nevertheless, the female reciter was assured to be included in the training subset. See Table 3 for a summary of the samples' distribution across resulting dataset splits.

B. TARTEEL DATASET

Tarteel is a publicly available dataset of Quran recitations recorded from normal Muslims around the world in real environments. It has a high diversity in reciters and recitation styles, proficiency, and speed. Tarteel v1 in [37] was released

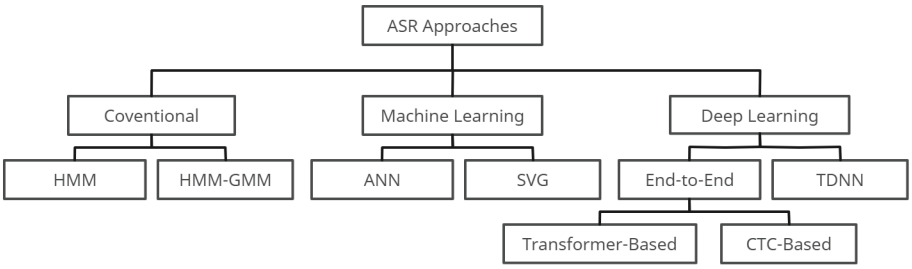


FIGURE 1. Summary of existing Quranic ASR approaches.

TABLE 2. Example of EqraTech database files content for one utterance (relations between parts are expressed using underlined text styles).

Database File	Line No.	Content
*.fileids	52301	abdusamad/019003
*.transcription	52301	<s> خَفِيًّا (1) نِدَآءً (2) إِذْ نَادَىٰ رَبَّهُۥ </s> (abdusamad/019003)
*.dic	-	نِدَآءً نِ دَ ا ا ا ا ء ن نِدَآءً (1) نِ دَ ا ا ا ا ء ن

TABLE 3. Labeled EqraTech dataset splits and distribution of samples.

Dataset Split	Hours	Samples	Speakers
Train	447.15	218258	68
Dev	52.18	23762	11
Test	48.14	25174	11

at the beginning of 2019 containing 20.5K recordings. The dataset provides demographic information of the reciters, such as age, sex, heritage, and Qira'ah. Each sample has Surah and Ayah numbers indicating what was recited. However, some recitations were found to be wrongly labeled or incomplete (do not cover the entire labeled Ayah). Notably, many recordings involved repetitions, corrections, hesitations, etc.

To focus on females and young ages, we proposed to extract subsets from the dataset, namely: *Female*, *13y*, and *13-18y*. First of all, we reformatted the dataset in a similar manner to the EqraTech dataset. Exceptionally, Tarteel did not provide direct transcriptions. Therefore, we used Ayah and Surah codes to produce word transcriptions, with the help of the Uthmani text of the Quran from Tanzil. Using only samples auto-evaluated as correct, the three resulting subsets follow in Table 4.

TABLE 4. Tarteel dataset splits and distribution of samples.

Type	Hours	Samples
Female	1.57	498
13y	0.2	70
13-18y	0.74	300

IV. METHODOLOGY

A. PHONEMES LIST

This work involved phonemes representing standard Arabic consonants and vowels in addition to Tajweed special phonemes. As Table 5 indicates, we considered 53 phonemes (including additional *Space*).

Classical Arabic has 28 consonants, 3 short vowels (diacritic signs), and their 3 long vowels [38]. However, some Arabic phonemes can be pronounced with one or more variants instead of the standard pronunciation in Quran recitation. Our phonemes cover many Tajweed rules of basic and advanced levels, including Qalqalah, Tafkheem, Consonant NOON and Tanween, Consonant MEEM rules, Ghunna, and Madd. The definitions of these rules can be found in [12]. In contrast, we suggested a few other phonemes to reflect common Tajweed mistakes regardless of Tajweed rules. To easily distinguish Quranic phonemes from others and keep the readability at the same time, we used Arabic-like symbols.

B. MOZILLA DeepSpeech ARCHITECTURE

Mozilla DeepSpeech is an open-source project that aims to provide commercial-quality speech recognition services to the public. It is actively developed by the contributions of a massive community of developers and researchers. The project offers easy-to-use Tensorflow implementation of Baidu's DeepSpeech in [8]. For these reasons, we proposed to use it in this work. Specifically, we used the 0.5.1 release of the project [39].

TABLE 5. List of phonemes used in training.

Phoneme	Description	Phoneme	Description
ء	HAMZAH, all forms		
ا	ALEF letter (long vowel)	آ	ALEF TAFKHEEM
ب	BEH letter	پ	BEH QALQALAH
ت	TEH letter, TEH_MARBUTA (continuing)		
ث	THEH letter		
ج	JEEM letter	چ	JEEM QALQALAH
ح	HAH letter		
خ	KHAH letter	خ	KHAH TAFKHEEM
د	DAL letter	ڈ	DAL QALQALAH
ذ	THAL letter		
ر	REH letter	ر	REH TAFKHEEM
ز	ZAIN letter		
س	SEEN letter		
ش	SHEEN letter		
ص	SAD letter	ص	SAD TAFKHEEM
ض	DAD letter	ض	DAD TAFKHEEM
ط	TAH letter	گ	TAH TAFKHEEM
		گ	TAH QALQALAH
ظ	ZAH letter	ظ	ZAH TAFKHEEM
ع	AIN letter		
غ	GHAIN TAFKHEEM letter		
ف	FEH letter		
ق	QAF letter	ق	QAF QALQALAH
		ق	QAF TAFKHEEM
ك	KAF letter		
ل	LAM letter	ل	LAM TAFKHEEM
م	MEEM letter	م	MEEM GHUNNA
ن	NOON letter	ن	NOON GHUNNA
		ن	NOON IKHFA'1
		ن	NOON IKHFA'2
ه	HEH letter, TEH_MARBUTA(stopping)		
و	WAW letter, includes long vowel	و	WAW GHUNNA
ي	YEH letter, includes long vowel	ي	YEH GHUNNA
◌َ	FATHA short vowel		
◌ُ	DAMMA short vowel		
◌ِ	KASRA short vowel		

DeepSpeech is composed of two phases: an acoustic model followed by a CTC decoder. The acoustic model consists of a multi-layer Recurrent Neural Network (RNN) with three fully connected layers, followed by a recurrent layer, and a fully connected layer before the final softmax output layer. Each non-recurrent hidden layer is followed by a clipped Rectified Linear Unit (ReLU) activation. Compared to Baidu's architecture, Mozilla replaced the bidirectional recurrent cells with unidirectional Long Short-Term Memory (LSTM) for long-term dependencies. The network is trained using CTC loss to map transcriptions to audio frames without forced alignment.

In the feed-forward process, the input audio is divided into overlapping windows of equal sizes. For each time-slice window, an MFCC feature vector is extracted and fed into the network. The final output at each time step is a vector of phoneme probabilities (for our phonemes in addition to the blank). The CTC decoder is responsible for transforming the resulting probabilities into textual transcripts. Mozilla uses the Adam method (which stands for adaptive moment

estimation) for network optimization. The overall structure is illustrated in Figure 2.

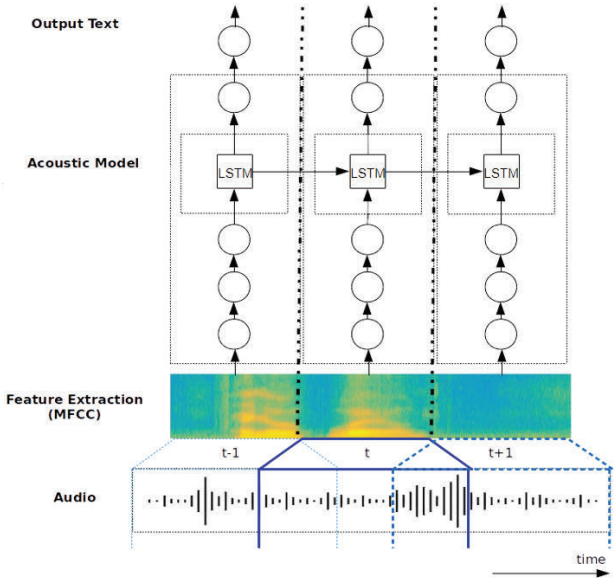


FIGURE 2. Structure of mozilla's DeepSpeech model shown unfolded in time for three time steps.

C. SELF-TRAINING PSEUDO-LABELING

The amount of our unlabeled EqraTech data is approximately triple when compared to the labeled records. It is a dataset full of real-world recitations collected from various groups of reciters. Therefore, we proposed using a semi-supervised method called Self-Training to expand our training dataset with these rich data.

Self-training generates pseudo-labels on the unlabeled data using a model trained on labeled ones. The accuracy of self-training depends largely on generating high-confidence pseudo-labels. Thus, we introduced data selection based on pseudo-labels matching between our model and an HMM-based recognizer. We used the model trained using the CMU Sphinx−4 toolkit by [18] for this purpose. We supported the HMM model with small language models for each page of the Holy Quran. Given the page number information for recitations, this makes the HMM model more reliable. Then, we used the WER measure on the Imla'ei predictions to calculate the confidence score. Finally, we selected interesting samples that exactly match with 100% confidence score. Figure 3 illustrates our self-training pseudo-labeling method.

Unlabeled data contains recitations of different proficiency levels. Therefore, we decoded our results with a language model that retains the recitation errors, as much as possible. Self-training is an iterative approach. Since we considered high-confidence labels only, we limited pseudo-labeling to one iteration.

D. TRANSFER LEARNING AND MODEL TRAINING

DeepSpeech is typically trained in a supervised manner, using input audios and their desired output labels. In this work, we proposed to start with supervised training as the baseline. Whilst, we developed the final model using transfer learning followed by semi-supervised training. We fed the training data into DeepSpeech in batches and sorted order by audio duration. Model parameters were adjusted with iterative optimization to minimize the CTC error between the predicted output and the actual label. During training, we applied dropout regularization after the non-recurrent layers to reduce overfitting and improving the generalization.

Mozilla provided an American English pre-trained model in their project. It was trained on the public Fisher, Librispeech and SwitchBoard datasets. It achieved a high accuracy of 8.22% WER on the LibriSpeech clean test dataset. We proposed to apply Transfer Learning to take advantage of this English pre-trained model. Despite that the pre-trained model is for a different language, we assumed it has useful noise and speech knowledge without gender bias.

In transfer learning, the output layer was replaced with our new softmax layer relevant to our Quranic phonemes. Then, we fine-tuned all model layers using the labeled EqraTech train dataset. For the final model, we trained our transferred model using the labeled EqraTech train dataset in combination with the high-confidence pseudo-labeled dataset. Figure 4 summarizes the overall training methodology.

In Table 6, we summarize the comparison between the training methods used in this work.

E. DECODING AND LANGUAGE MODELS

DeepSpeech supports the use of an external n-gram language model (LM) to improve the accuracy of the decoded transcripts. The text corpus used in building the language model has the main impact on its effectiveness. In this work, we proposed building multiple LMs to handle various applications, as follows:

1) ***EqraTech Transcriptions LM*** for model basic evaluation purposes. The text corpus consisted of the phonemic transcriptions of the clean labeled EqraTech dataset.
2) ***Phonemic Isolated Words LM*** to retain common mistakes of Tajweed in isolated words. The LM text corpus involved all correct phonemic pronunciation variants included in our pronunciation dictionary. Additionally, we added several erroneous variants to support common mistakes of non-native Arabic speakers or regional accents. We extracted them by applying defined rules of alternating some phonemes on the original dictionary. For instance, it is common to mispronounce DAD as a ZAH in Iraq or replace a SEEN with a ZAIN and vice versa in Egypt, etc.
3) ***Phonemic Continuous Ayahs LM*** to cover the entire Quran with considering continuous recitation. The

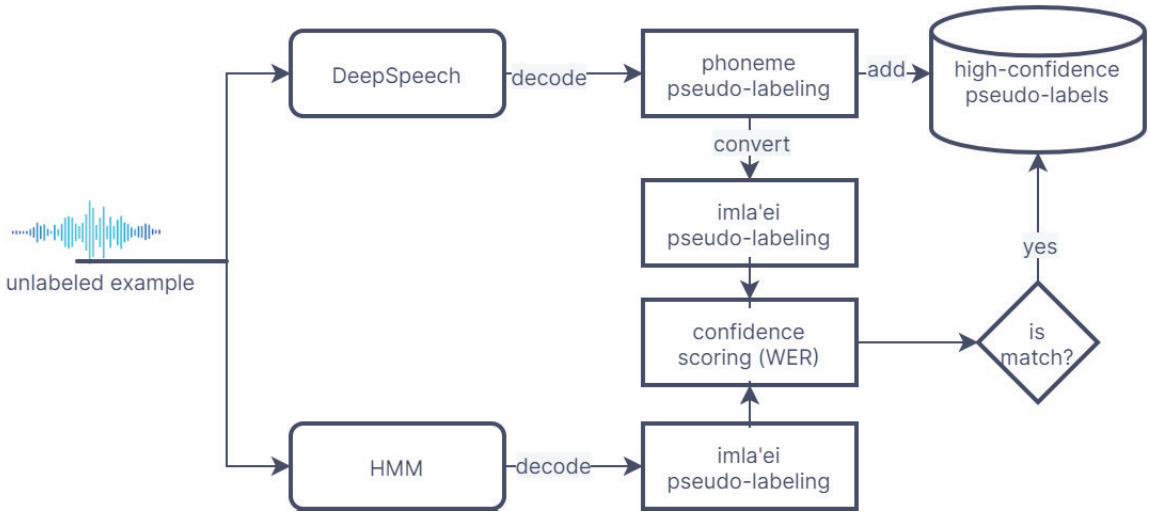


FIGURE 3. Illustration of DeepSpeech self-training pseudo-labeling.

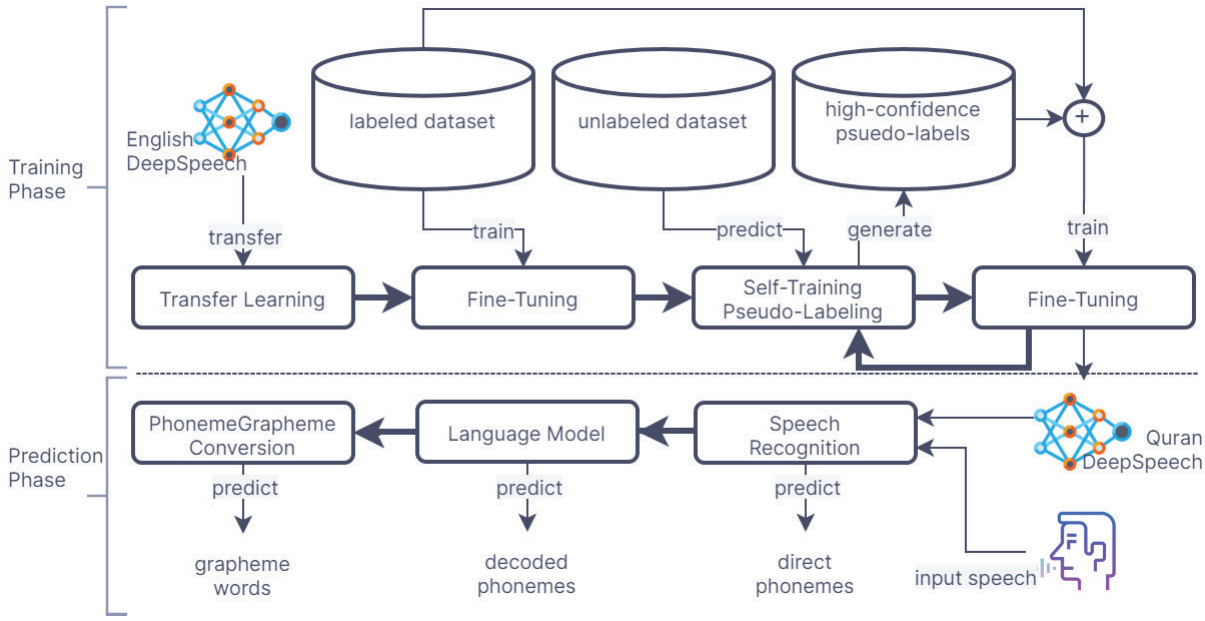


FIGURE 4. Overall methodology of our transfer learning and semi-supervised hybrid model.

text corpus consisted of eight valuable phonemic versions of the Quran text. Each version corresponds to a common correct or erroneous Tajweed style. We generated the corpus using automatic Tajweed rules in [18] with the help of Tanzil Uthmani text. The LM corpus included isolated Ayahs in addition to connected Ayahs (connecting the last two words of one Ayah with the next Ayah).

We used the latest stable release of the KenLM tool to generate our language models [40].

DeepSpeech uses beam search decoding to get the best results according to both the RNN output and the used language model. The decoding objective function is a weighted combination of the LM score and RNN score, with α representing the language model weight and β for the word insertion weight. That is, the optimal α and β coefficients lead to the best recognition results.

F. PHONEME-TO-GRAPHEME CONVERSION

Our model predicts transcriptions at the phoneme level. However, in some applications, transcription results at the word level (grapheme) are more useful. We apply such conversions as a final step in our system, post-processing. Using a simple reverse look-up of our Uthmani pronunciation dictionary has its limitations. Many words have redundant occurrences in the dictionary with slight writing differences.

TABLE 6. Comparison summary between used training methods.

	Supervised Learning	Transfer Learning	Self-Training
Training Process	Trained from scratch (random initial weights)	Fine-tuning a pre-trained model	Supervised training followed by self-training on unlabeled data
Data Requirements	labeled data	Pre-trained model and labeled data	Initial labeled data, then self-generated pseudo-labels
Unlabeled Data Usage	No	No	Yes
Use Cases	Sufficient labeled data for each class of interest is available	Related domain knowledge is available	Insufficient labeled data or imbalanced dataset
Risks	Poor generalization with limited labeled data	Limited relevance between pre-trained and target domains	Introducing errors with self-generated labels
Algorithm Complexity	Simple as it learns from data directly	Moderate as it involves transfer and freezing layer selection	Complex as it involves additional labeling, selection and retraining

Some Uthmani words correspond to merged words. Indeed, a sequence of connected phonemes may correspond to numerous possible different combinations of words.

As a solution, we suggested having our words in the simple Imla'ei format. Foremost, we normalized each word using a Uthmani-to-Imla'ei mapping. We were able to semi-automatically generate a one-to-one map making use of two digital Quran text versions from Tanzil: Uthmani and simple Imla'ei. Finally, we used a language model to score the different combinations and obtain the best among them. This graphemic Quran LM was trained on the entire Imla'ei Tanzil Quran text as a continuous text.

G. EVALUATION METRICS FOR EVALUATING TEXT TRANSCRIPTION

WER is the most common metric for evaluating continuous speech recognition systems [9]. It calculates the minimum edits needed for a hypothesis (i.e., predicted) transcription to match a reference (i.e., the ground truth). WER's formula for graphemic Imla'ei words is shown in Equation 1. It sums up the three types of edits (substitution, deletion, and insertion), over the length of the reference string.

$$WER = \frac{S+D+I}{N} \times 100\% = \frac{S+D+I}{S+D+C} \times 100\%, \quad (1)$$

where, S, D, and I are the numbers of words substituted, deleted or inserted, respectively, in order for the predicted transcript to match the reference, N is the number of words in the reference, and C is the number of correct words. WPER and PER are defined similarly but using phonemic word, and phonemes, respectively.

We used an additional metric ($\text{WER}_{without_deletions}$) for system performance evaluation purposes. $\text{WER}_{without_deletions}$ discards inaccurate source transcriptions, such as incomplete recited Ayah which is labeled with the complete Ayah transcription. This yields to Equation 2, as follows:

$$WER_{(without_deletions)} = \frac{S+I}{S+C} \times 100\% \quad (2)$$

The work methodology corresponds to the design of an effective speech recognition system that can assist widely with the Quran recitation of normal Muslims all over the world. The design of our system is capable of identifying phonemes from very basic Arabic to advanced Quran Tajweed rules. DeepSpeech is a robust model that conducts end-to-end speech recognition with handling noisy environments. The proposed design trains DeepSpeech in a two-stage methodology to effectively support both male and female Quran reciters, with no age limitation. Finally, by having multiple language models and transcription evaluation techniques, the system can be used in a wide range of digital Quran applications concerning either phoneme or word-level recognition. We illustrate the overall methodology in Figure 4.

V. RESULTS AND DISCUSSIONS

A. EXPERIMENTAL SETUP

All experiments in this work were conducted on a laptop computer running a 64-bit Ubuntu Linux 16.04 LTS operating system. It has an Intel® Core™ i7-7700HQ 2.80GHz CPU with 8 cores and 16G RAM. It also has a single NVIDIA GeForce GTX 1070 GPU with 2048 CUDA Cores and 8 GB total memory.

We used Mozilla's DeepSpeech v0.5.1 as our core implementation. We fixed the model parameters to the values summarized as in Tables 7 for final experiments. We used the EqraTech *dev* for validation during development and *test* for basic evaluation. We used *Female, 13y and 13-18y* Tarteel subsets for further evaluations concerning women's and children's speech.

B. DECODING AND LANGUAGE MODELS EXPERIMENTS

We have done experiments to study our model's decoding results with various language models. Here, we used a model trained with supervised learning on the EqraTech train subset. The initial weights of the model were randomly chosen using the default initializer of Tensorflow. With decoding parameters (α, β, *beam_width*) set to (0.75,1.85,1024), Table 8 summarizes the results on the EqraTech dataset.

Using LMs improved the results noticeably. Compared to EqraTech transcriptions LM, larger language models

TABLE 7. Summary of model hyperparameters.

Hyperparameter	Value
# of MFCC features	26
Length of audio window (ms)	32
Step of audio window (ms)	20
Sample rate (Hz)	16000
Length of audio window (samples)	512
Step of audio window (samples)	320

(a) audio

Hyperparameter	Value
# of hidden neurons	2048
# of output neurons	54
ReLU clipping value	20.0
Learning rate	0.0001
Dropout rate	0.15
Train batch size	28
Development batch size	48

(b) training

Hyperparameter	Value
Test batch size	48
LM decoding α	1.0
LM decoding β	5.5
Beam width	1024

(c) testing

TABLE 8. Comparison with respect to various language models.

Language Model	Order	WPER	PER
Without LM		59.36	14.15
EqraTech transcriptions	6-gram	**29.25**	**07.62**
Phonemic Isolated Words	3-gram	48.89	12.54
Phonemic Continuous Ayahs	6-gram	**32.67**	**08.67**
	5-gram	32.70	08.67
	4-gram	32.77	08.68
	3-gram	33.00	08.72

negatively impacted the test results. Despite this, they are more suitable for real-world recognition, covering a wider range of recitation variations of the entire Quran. We also noticed that increasing the n-gram order did not help much.

Many errors in the direct results were due to the misidentifying of the space character between words. The space character is a delimiter between words without a special phonetic sound. It may represent a pause but, otherwise, it does not refer to any kind of alignment with the input speech. In Quran recitation, many space-separated words are merged while recited. This causes out-of-vocabulary words. Besides the space issue, many WPER and PER errors occurred because some phonemes are phonetically very close. The integration of our language models solved some of such problems on the word level.

To illustrate the importance of language models and the different result formats of our system, an example in Table 9 follows. Comparing the source to the direct result of this example, our model recognized GHAIN-TAFKHEEM(غ) QAF-TAFKHEEM(ڤ) instead. The two phonemes are similar in their sound but they do not relate to the same character, causing a clear Tajweed mistake. In contrast, the model predicted REH(ر) instead of REH-TAFKHEEM(ڑ) which corresponds to the same Arabic letter but with different phonetic characteristics. The latter is a hidden Tajweed mistake. Likewise, our model detected the WAW(و) phoneme to be stretched for two repeated counts instead of four in the source transcript. Stretching (MADD) is a Tajweed pronunciation characteristic where the count matters for recitation evaluation and correction. Next, the language model improvement appears in the decoded result. The language model mainly works as a spell checker that increases the likelihood of generating a valid Quran transcription. Finally, grapheme-to-phoneme conversion to Imla'ei format dissolves the errors. This stresses the fact that our model correctly recognized the utterance under analysis on the word level, despite the phonetic mispronunciation.

TABLE 9. Example of different result formats the model produces.

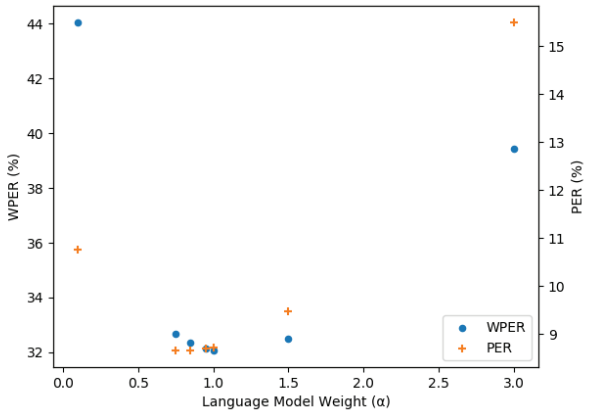
Format		Transcription
Phonetic	source:	غُلِبَتِ ڑڤوووم
	result (direct):	قُلِ بَتِ رڤوم
	result (decoded):	غُلِبَتِ ڑڤوم
Imla'ei	source:	غُلِبَتِ ٱلرُّومُ
	result:	غُلِبَتِ ٱلرُّومُ

PaddlePaddle's grid search in [41] shows the error surface with α and β variations as a convex function of α. According to this, we have done two experiments to find the optimal decoding parameters regarding our Quranic problem. In the first experiment, β and *beam_width* were fixed at 1.85 and 1024, and binary search results for α were found as shown in Figure 5a. In the second experiment, α was fixed at the optimal value of 1.0 and the error surface according to β variation was observed and illustrated in Figure 5b. Our optimal values, which produced a minimal WPER of 31.67% and 8.3% PER using our phonemic Ayahs LM, are listed in Table 7c.

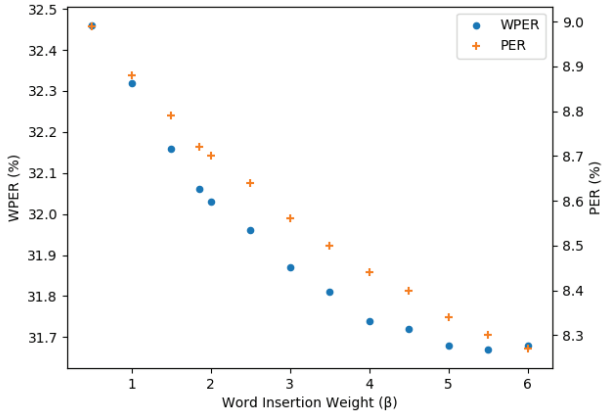
C. MODEL TRAINING EXPERIMENTS

First, we have evaluated our baseline model performance using the optimal decoding parameters. We trained the model from scratch with supervised learning using the EqraTech train subset. We used the default initializer of Tensorflow for the initial weights. Table 10 and Table 11 report the results over the EqraTech and Tarteel datasets.

The promising results of our baseline model validate adopting DeepSpeech on our phoneme-based research problem and our training data. This agrees with the good performance of using the DeepSpeech architecture over other languages, mentioning English, German, and Russian in [39], [42], and [43].



(a) $beam_width = 1024$, $\beta = 1.85$



(b) $beam_width = 1024$, $\alpha = 1.0$

FIGURE 5. Relationship between error rate and decoding hyperparameters.

For transfer learning, we have experimented with two fine-tuning strategies. The objective of this experiment is to study the transfer learning technique and evaluate the results compared to typical supervised learning. We first attempted freezing the first three fully-connected layers while fine-tuning the rest. We noticed that this technique accelerated the training process since fewer back-propagation gradient calculations were required. However, it failed to improve the results. Anyway, it was able to predict recitations with 18.8%PER, which is far from randomness. This supports the insight that the very first layers may be responsible for extracting general or low-level speech features that are not related to language patterns.

On the other hand, we tested fine-tuning all pre-trained model layers without any freezing. Table 10 summarizes the results on the EqraTech test subset. We found that it outperforms the model trained from scratch in nearly all aspects. However, it had a very slight increase in the phonetic error rates (PERs) while using language models. Table 11 summarizes the results of the Tarteel dataset. The reported improvements on Tarteel were significantly higher, compared to the EqraTech dataset. Specifically, the transfer-based model outperformed baseline results by 7−13% WER.

Our labeled training dataset is constrained to male adults who recite with high proficiency in a studio-quality recording environment. The obtained results confirmed the usefulness of transferring noise and speech knowledge from the well-trained English model to the Quran. However, improvements on the females' samples were the highest noticed. This may relate to the fact that the English model was trained on adult speech data.

Considering the transfer learning-based model as a seed model for self-training, we were able to generate high-confidence pseudo-labels in just one iteration. The new data (318,260 samples of 556.34 hours) was approximately one-third of the entire unlabeled data. The samples of this data were distributed over half the total speakers (27,854 reciters). The generated data was larger than the manually labeled EqraTech train dataset by 100K samples.

Finally, we have experimented with a hybrid training approach of transfer learning and semi-supervised. In this experiment, our objective is to study exploiting the newly generated data which is rich with real-world recitations recorded by diverse normal reciters all over the world. Table 10 and Table 11 show that semi-supervised training further enhanced the model performance compared to transfer learning alone. The final hybrid model is the best model obtained on all evaluation aspects.

For visual comparisons, the results presented in Tables 10 and 11 are plotted in Figures 6 and 7, respectively.

Regarding the EqraTech dataset, the Imla'ei result was highly boosted to reach 3.11%WER and 6.18%PER. Some high error rates are due to low-quality annotation. For example, we found some samples that start recitation with Isti'adhah (استعاذة in Arabic) or Basmallah (بسملة in Arabic), which were missing in the transcription. For other samples, we noticed that computer-generated results were more accurate than the original transcription at the phoneme level, especially when distinguishing between similar phonemes. Discovering whether a mispronounced fault lies with the annotation or with the recognizer was difficult.

Compared to the baseline model, our best model has remarkable improvements on the in-the-wild Tarteel dataset. For female recitations, our model improved the Imla'ei word results from 23% WER. Likewise, recognition of the young ages improved by 18-23% WER and 15-21% $WER_{without_deletions}$. Some of these errors are due to speech repetitions found in Tarteel recitations. The results of the WER metric were less significant compared to the $WER_{without_deletions}$ metric. This is somehow related to imperfect transcriptions such as incomplete recited Ayahs. Although Tarteel subsets are considered small for reliable evaluation, the results give a good indication of improvements.

TABLE 10. Summary of the results on EqraTech test dataset.

Training Method	Without LM		Transcriptions LM			Phonemic Ayahs LM		
	WPER	PER	WPER	PER	WER	WPER	PER	WER
Supervised Learning	59.36	14.15	28.31	7.23	5.22	31.67	8.3	6.46
Transfer Learning	55.35	13.67	26.54	7.3	4.24	29.76	8.66	5.14
Transfer Learning+ Semi-supervised	**52.94**	**12.24**	**24.9**	**6.18**	**3.11**	**27.97**	**7.4**	**3.92**

TABLE 11. Summary of the results on Tarteel dataset subsets.

Training Method	Female		13y		13-18y	
	WER	WER $w/o\ D$	WER	WER $w/o\ D$	WER	WER $w/o\ D$
Supervised Learning	63.43	48.72	44.17	33.55	68.19	53.54
Transfer Learning	50.10	32.49	37.53	23.17	58.57	40.31
Transfer Learning+Semi-supervised	**40.43**	**25.39**	**31.17**	**17.93**	**50.60**	**32.13**

TABLE 12. Comparative study between the presented work and other existing works in the literature.

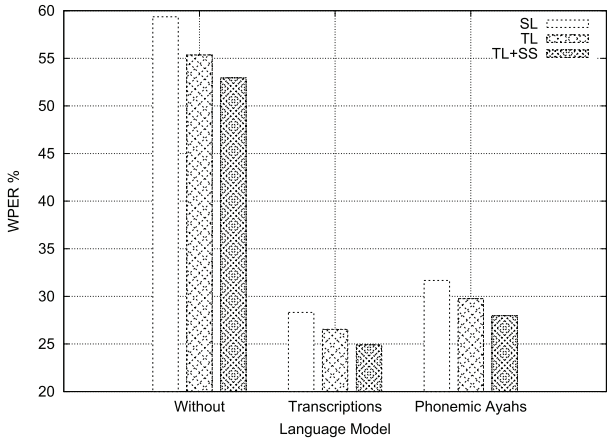
Work	Dataset			Methodology					
	#Verses	#Suras	#Reciters	DL-based	LM	AM	Template	Traditional	E2E
[29]	∈[101,200]	1	32	✓	✓	✓		✓	
[44]	N	N	13	✓	✓	✓		✓	
[45]	>200	N	10	✓			✓		
[46]	>200	49	39		✓	✓		✓	
[47]	>200	3	4				✓		
[48]	∈[1,100]	N	10				✓		
[49]	>200	N	7	✓			✓		
[50]	∈[1,100]	N	60	✓	✓	✓		✓	
[51]	N	N	60	✓	✓	✓			✓
[30]	>200	37	42	✓		✓			✓
[52]	>200	N	129	✓		✓			✓
[53]	N	N	N	✓			✓		
[54]	>200	114	35	✓	✓	✓			✓
This work supervised	>200	114	68	✓	✓	✓			✓
This work semi-supervised	>200	114	68 + 27854	✓	✓	✓			✓

Note: N means not specified.

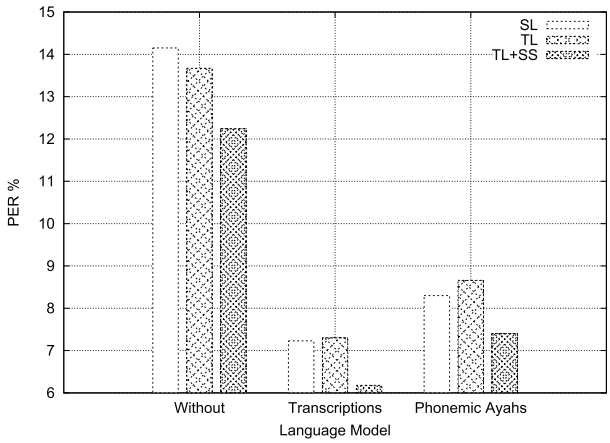
TABLE 13. Comparisons with those works, in the literature, that use the E2E learning techniques.

Work	Dataset			Architicture	WER
	Quran verses%	#Clip files	Audio duration (Hours)		
[51]	1.6%	960	10	Transformer	6.16%
[30]	9%	16207	≈ 50	CNN-Bidirectional GRU - CTC	8.34%
				TDNN-CTC	45.73%
[52]	N	72735	>100	RNN-CTC	19.43%
				Transformer	95.03%
[54]	100%	217121	2698	BLSTMP-CTC/Attention	10.4%
This work supervised	100%	218258	447.15	RNN-CTC	4.24%
This work semi-supervised	100%	218258+318260	447.15+556.34	RNN-CTC	3.11%

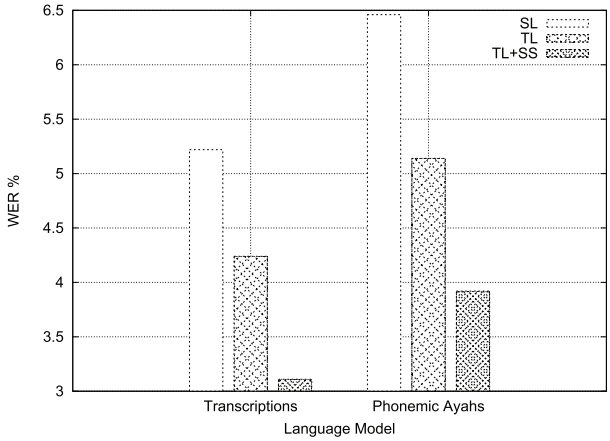
We have found that the training method has a great impact on the required training time. All models were trained for a maximum of 15 epochs using a batch size of 28. For the baseline model and transfer learning-based model which used 218K training data, one epoch corresponded to 7,794 learning steps. In contrast, the semi-supervised model had 19,161 learning steps per epoch. We needed approximately 4, 6, and 15 hours for one training epoch to finish using supervised



(a) WPER

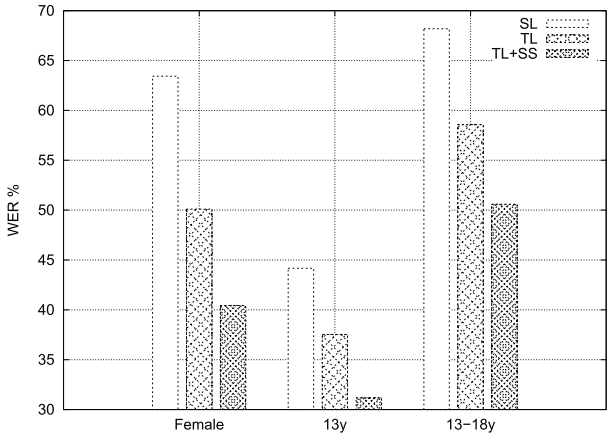


(b) PER

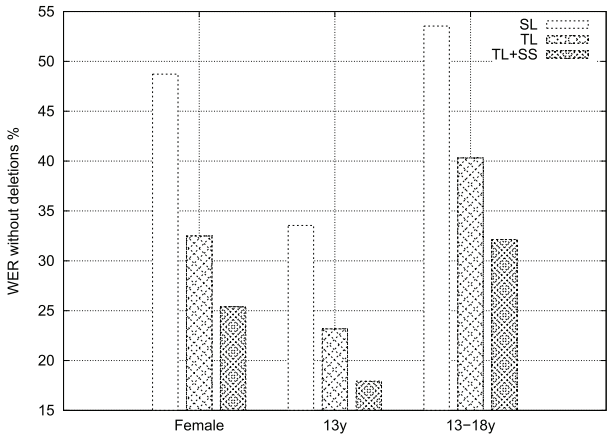


(c) WER

FIGURE 6. **Visual comparison of the results based on EqraTech dataset.**



(a) WER



(b) WER without deletions

FIGURE 7. **Visual comparison of the results based on Tarteel dataset.**

learning, transfer learning, and semi-supervised training, respectively. Notably, our best model was the slowest in epoch time due to the larger training data used. Despite this, it required much fewer training epochs to converge than training from scratch. The best validation (dev) performance occurred after 14, 6, and 4 epochs for supervised learning, transfer learning, and semi-supervised training, respectively.

Comparisons between the presented work and other existing works in the literature are given in Table 12.[6] These comparisons are made based on the characteristics of the dataset and the adopted methodology. The characteristics of the dataset include number of investigated verses, number of suras, and number of different reciters. For the methodology part, the comparison is made based on whether the work uses or does not use: deep learning (DL), language model (LM), acoustic model (AM), template-based speech recognition techniques, traditional-based speech recognition techniques, and end-to-end (E2E) learning techniques. For the semi-supervised version of this work, number of reciters is presented as (68 + 27854). The 68 represents specific well know reciters. While, the 27854 represents anonymous reciters who are, in fact, users who use the application.

Moreover, the presented work is further compared with those works, in the literature, that use the E2E learning techniques. As shown by Table 13,[7] in this case, the comparison is made based on the characteristics of the dataset, the architecture, and the WER. The characteristics of the dataset that are used are percent of the Quran verses,

[6]Part of the content of this table is adapted from [54].
[7]Part of the content of this table is adapted from [54].

number of clip files, and the audio duration in hours. For the semi-supervised version of our work the number of clip files is presented as (218258+318260). The 218258 clip files are for specified well known reciters. While the 318260 clip files are for anonymous users of the application. Best WER of 3.11% is achieved by our semi-supervised version of our work.

Generally speaking, our innovative solution leverages different techniques to produce a high-performing system compared with existing ones. For example, using transfer learning allows for leveraging general acoustic knowledge. Pretraining on a massive English dataset allows the model to learn highly generalizable, language-agnostic acoustic features. This provides a strong ''understanding'' of human speech. Another advantage is improved initialization and faster convergence. Starting from these pretrained weights provides a superior initial state compared to random initialization, significantly accelerating convergence during fine-tuning on the smaller Arabic dataset and helping prevent overfitting. Finally, using transfer learning can lead to enhanced feature extraction and robustness. The pretrained acoustic encoder acts as an effective feature extractor, improving the Arabic model's ability to generalize and enhancing its robustness across diverse real-world audio environments [55], [56], [57], [58].

As for the use of semi-supervised learning, it is critical for further enhancing performance by exploiting the vast amount of available unlabeled Arabic speech data. This approach overcomes the bottleneck of costly manual transcription by effectively utilizing large quantities of unlabeled data, which is especially beneficial for Arabic. Moreover, by using the self-training process the training set is significantly expanded, allowing the model to learn from a much larger corpus. Finally, learning from this extensive unlabeled Arabic data refines the model's understanding of Arabic-specific acoustic characteristics and linguistic patterns, leading to improved generalization and adaptation to real-world Arabic speech [59], [60].

In conclusion, the superior performance of our system is a direct result of the synergistic combination of DeepSpeech's efficient end-to-end architecture, the knowledge transfer from a high-resource language through pretraining, and the scale-up achieved by effectively leveraging abundant unlabeled data via semi-supervised learning. This multi-pronged approach enables us to build a high-performing ASR system for a low-resource, morphologically rich language like Arabic, pushing the boundaries of what is achievable with limited labeled data.

VI. CONCLUSION AND FUTURE WORK

In this paper, we have presented a DeepSpeech-based speech recognition system for the Holy Quran. The model converts real users' recitations into potential phonemes being spoken, and, optionally from these phonemes to possible words. Specifically, the model can identify 52 phonemes of standard Arabic and major Tajweed rules. Our proposed model offers Quran learning assistance by providing live feedback for correcting recitation mistakes. DeepSpeech can be easily adapted to the Quran, achieving reasonable results with few training epochs only. The system tackled the absence of balanced labeled datasets through proposed training strategies. Surprisingly, fine-tuning an available pre-trained English model using transfer learning has boosted our model performance. Our final model has been evaluated on a private EqraTech dataset decoded with a real-world language model. Evaluations showed 7.4%PER and 27.97%WPER at the phoneme level and 3.92% WER at the Imla'ei word level. The EqraTech test subset exclusively contains male reciters, making it less reliable for performance evaluation of females and children. We have, thus, conducted evaluations on the Tarteel dataset to show the effectiveness of our model. We have provided the best-obtained model along with the supporting language models to EqraTech company to deploy them in their Quranic digital products.

Despite the importance of our work, it is still imperfect with many areas for improvement. For example, we could not make any benchmark comparison to existing models in the literature due to the scarcity of publicly available Arabic and Quranic datasets. We suggested evaluating the public Tarteel dataset as a step toward comparison with future research. However, female and young age subsets are small datasets with non-phonetic transcriptions that are inaccurate for several samples. This paper highlighted the need for well-annotated public datasets for testing concerning different genders and ages.

Public balanced datasets that have been phonetically transcribed are required for training in Arabic ASR research. In this work, we were not able to publish the used dataset or the trained model for privacy reasons. However, we tend to work in open research in the future to support the field of speech recognition research concerning the Holy Quran and the Arabic language. Future directions for this research should include more investigation to tackle the possible gender bias introduced by unbalanced speech data used.

The work done here focused on basic Tajweed rules. Future work can include all advanced Tajweed rules to make the system suitable for Quran learners at all levels. Other areas to consider include handling common yet extra challenging settings where the reciters are reciting the verses at a very high tempo (while trying to memorize it).

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